Ajay Kumar Jha

Curriculum Vitae

QBB 258 A23, Department of Computer Science
North Dakota State University
Fargo, ND 58105, USA
ajay.jha.1@ndsu.edu ☎ *(701) 231-6377*
https://hifromajay.github.io/

Research Interests

Software Engineering, Software Testing and Maintenance, Mining Software Repositories, and Empirical Software Engineering.

Professional Experience

Assistant Professor (Aug 2022 – Present)
Department of Computer Science, North Dakota State University, Fargo, ND, USA

Postdoctoral Researcher (Mar 2020 – Jul 2022)
University of Alberta, Edmonton, Alberta, Canada

Postdoctoral Researcher (Mar 2017 – Feb 2020)
Kyungpook National University, Daegu, Republic of Korea

Co-founder (Feb 2009 – Jun 2011)
Unified Technology Consultancy, Kathmandu, Nepal

Account Manager (Mar 2008 – Aug 2008)
V2Soft, Bangalore, India

Business Development Manager (Sep 2007 – Jan 2008)
FoxsysTech, Gurgaon, India

Co-founder (Nov 2005 – Aug 2007)
MindproSoft, Bangalore, India

Education

Ph.D. in Computer Science and Engineering, Feb 2017
Kyungpook National University, Republic of Korea

M.S. in Computer Science, Aug 2013
Kyungpook National University, Republic of Korea

B.Sc. in Information Technology, Jan 2004
Sikkim Manipal University of Health, Medical and Technological Sciences, India

Research Funding

NSF CAREER Award (Jan 2027 – Dec 2031), PI, $500,000
CAREER: Improving Software Test Quality and Developer Skills through Developer-in-the-Loop Test Optimization

USDA NIFA (May 2026 – May 2029), Co-PI, $591,500
A-SYSRV: Systems Biology Reverse Vaccinology, An Integrated Vaccine Prediction Tool for Aquaculture Pathogens
Contribution: Web application development

USAID (Cancelled), Co-PI, $481,225
Digital Tools for Optimal Fish Feed Formulation in Nigeria
Contribution: Mobile app development

ND EPSCoR (Oct 2022 – Jun 2024), PI, $20,000
Unit Test Recommendation and Reuse

Publications

- David Owuor, **Ajay Kumar Jha**. Developer vs. DSpot vs. ChatGPT: A Comparative Study of JUnit Test Amplification. In *19th IEEE International Conference on Software Testing, Verification and Validation (ICST)*, 2026. SVE Track.
- Mohayeminul Islam, **Ajay Kumar Jha**, May Mahmoud, Ildar Akhmetov, and Sarah Nadi. An Empirical Study of Python Library Migration Using Large Language Models. In *Proceedings of 40th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, pp. 867-879, 2025.
- Suraj Bhatta, Frank Kendemah, **Ajay Kumar Jha**. Understanding Test Deletion in Java Applications. In *Proceedings of 22nd IEEE/ACM International Conference on Mining Software Repositories (MSR)*. pp. 408-420. 2025.
- **Ajay Kumar Jha**, Sarah Nadi. Migrating Unit Tests Across Java Applications. In *Proceedings of the 24th IEEE International Conference on Source Code Analysis and Manipulation (SCAM)*. pp. 131-142. 2024.
- Sai Kiran Bhrugumalla, **Ajay Kumar Jha**. TRec: A Regression Test Recommender for Java Projects. In *Proceedings of 40th IEEE International Conference on Software Maintenance and Evolution (ICSME)*. pp. 903-907. 2024. Tool Demo Track
- Mohayeminul Islam, **Ajay Kumar Jha**, Ildar Akhmetov, and Sarah Nadi. Characterizing Python Library Migrations. In *Proceedings of 32nd ACM Symposium on the Foundations of Software Engineering (FSE)*. pp. 92-114. 2024.
- Mohayeminul Islam, **Ajay Kumar Jha**, Sarah Nadi, and Ildar Akhmetov. PyMigBench: A Benchmark for Python Library Migration. In *Proceedings of 20th IEEE/ACM International Conference on Mining Software Repositories (MSR)*. pp. 511-515. 2023. Data and Tool Track.
- **Ajay Kumar Jha**, Mohayeminul Islam, and Sarah Nadi. JTestMigBench and JTestMigTax: A benchmark and taxonomy for unit test migration. In *Proceedings of 30th IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*. pp. 713-717. 2023. ERA Track.
- Mansur Gulami, **Ajay Kumar Jha**, Sarah Nadi, Karim Ali, Emily Jiang, and Yee-Kang Chang. A Human-in-the-loop Approach to Generate Annotation Usage Rules: A Case Study with MicroProfile. In *Proceedings of Annual International Conference on Computer Science and Software Engineering (CASCON)*, pp 91-100. 2022.
- Batyr Nuryyev, **Ajay Kumar Jha**, Sarah Nadi, Yee-Kang Chang, Emily Jiang, and Vijay Sundaresan. Mining Annotation Usage Rules: A Case Study with MicroProfile. In *Proceedings of 38th IEEE International Conference on Software Maintenance and Evolution (ICSME)*. pp. 553-562. 2022. Industry Track
- **Ajay Kumar Jha** and Sarah Nadi. Annotation practices in Android apps. In *Proceedings of 20th International Working Conference on Source Code Analysis and Manipulation (SCAM)*, pp. 132-142. IEEE, 2020.
- Sooyong Jeong, **Ajay Kumar Jha**, Youngsul Shin, and Woo Jin Lee. A log-based testing approach for detecting faults caused by incorrect assumptions about the environment. *IEICE Transactions on Information and Systems*, 103(1), pp. 170-173.
- **Ajay Kumar Jha**, Sunghee Lee, and Woo Jin Lee. An empirical study of configuration changes and adoption in Android apps. *Journal of Systems and Software (JSS)*. 156 (2019), pp. 164-180.
- **Ajay Kumar Jha**, Sunghee Lee, and Woo Jin Lee. Characterizing Android-specific crash bugs. In *Proceedings of the 6th IEEE/ACM International Conference on Mobile Software Engineering and Systems (MobileSoft)*, pp. 111-122. 2019.
- **Ajay Kumar Jha**, Deok Yeop Kim, and Woo Jin Lee. A framework for testing Android apps by reusing test cases. In *Proceedings of the 6th IEEE/ACM International Conference on Mobile Software Engineering and Systems (MobileSoft)*, pp. 20-24. 2019. Vision Track.
- **Ajay Kumar Jha** and Woo Jin Lee. An empirical study of collaborative model and its security risk in Android. *Journal of Systems and Software (JSS)*. 137 (2018), pp. 550-562.

- **Ajay Kumar Jha**, Sunghee Lee, and Woo Jin Lee. Developer mistakes in writing Android manifests: An empirical study of configuration errors. In *Proceedings of the 14th International Conference on Mining Software Repositories (MSR)*, pp. 25-36. 2017.
- **Ajay Kumar Jha** and Woo Jin Lee. Analysis of permission-based security in android through policy expert, developer, and end user perspectives. *Journal of Universal Computer Science (JUCS)*, vol. 22, no. 4, pp. 459-474, 2016.
- **Ajay Kumar Jha**, Sunghee Lee, and Woo Jin Lee. Modeling and test case generation of inter-component communication in android. In *Proceedings of the Second ACM International Conference on Mobile Software Engineering and Systems (MobileSoft)*, pp. 113-116. 2015.
- **Ajay Kumar Jha**, Seungmin Lee, and Woo Jin Lee. Permission-based security in android application: from policy expert to end user. In *Proceedings of the 2015 Research in Adaptive and Convergent Systems (RACS)*, pp. 319-320. ACM, 2015.
- Soo Young Jang, **Ajay Kumar Jha**, and Woo Jin Lee. Virtual prototype generation by shockwave flash for simulating HW components of embedded system. In *Proceedings of the 29th Annual ACM Symposium on Applied Computing (SAC)*, pp. 1755-1756. 2014.
- **Ajay Kumar Jha** and Woo Jin Lee. Activity-based Event Capture and Replay Technique for Reproducing Crashes in Android Applications. *IEMEK Journal of Embedded Systems and Applications*. 9 (1), 1-9. 2014
- **Ajay Kumar Jha**, Sooyong Jeong, and Woo Jin Lee. Value-deterministic search-based replay for android multithreaded applications. In *Proceedings of the 2013 Research in Adaptive and Convergent Systems (RACS)*, pp. 381-386. ACM, 2013.
- **Ajay Kumar Jha** and Woo Jin Lee. Capture and replay technique for reproducing crash in android applications. In *Proceedings of the 12th IASTED International Conference in Software Engineering*, pp. 783-790. 2013.

Student Supervision and Mentoring

Ph.D. Students Advised

- **Frank Kendermah**, North Dakota State University (Jan 2026 - Present)
- **Rupinder Kaur**, North Dakota State University (March 2024 – June 2025)
- **Amjad Allobadi**, North Dakota State University (Aug 2022 – Dec 2023)

Ph.D. Students Mentored

- **Mohayeminul Islam**, University of Alberta (Oct 2021 – Aug 2025)

Master's Students Advised

- **Chloe Chinault,** North Dakota State University (Sep 2025 - Present)
- **Aayush Lamsal**, North Dakota State University (Feb 2025 - Present)
- **Suraj Pokhrel**, North Dakota State University (Aug 2024 – Dec 2025)
- **David Owuor**, North Dakota State University (May 2024 – Dec 2025)
- **Frank Kendermah**, North Dakota State University (Feb 2024 – Dec 2025)
- **Suraj Bhatta**, North Dakota State University (Jan 2023 – Dec 2024)
- **Sai Kiran Bhrugumalla**, North Dakota State University (March 2023 – July 2024)

Undergraduate Research Interns

- **Griffin Winstead,** NDSU (Jan 2026 – May 2026), Supported by NDSU Explore Program (PI, $2,781)
- **Xichen Pan**, University of Alberta (May 2021 – Sep 2021)

Teaching Experience

North Dakota State University (*Newly developed course, ******Substantially revised course**)**

***CSCI 412:** Mobile Software Engineering, Fall 2023, 2024, 2025, 2026
***CSCI 712:** Mobile Software Engineering, Spring 2025, 2026
****CSCI 713:** Software Development Processes, Fall 2025, 2026

****CSCI 714:** Software Project Planning and Estimation, Fall 2022, 2023, 2024
***CSCI 783:** Topics in Software Systems, Spring 2023, 2024
****CSCI 790:** Graduate Seminar:
Optimizing Regression Test Suites – Spring 2024
LLMs for Software Testing and Maintenance – Fall 2024, Spring 2026
Code Smell and Refactoring – Spring 2025

Professional Service

Conference Session Chair

ICSME: International Conference on Software Maintenance and Evolution - 2024
SANER: International Conference on Software Analysis, Evolution and Reengineering – 2022
MSR: International Conference on Mining Software Repositories - 2022

Program Committee Member

MSR: International Conference on Mining Software Repositories (Technical track) – 2021, 2022, 2024, 2025, 2026, 2027
ICSME: International Conference on Software Maintenance and Evolution (Technical track) - 2024, 2025, 2026
SANER: International Conference on Software Analysis, Evolution and Reengineering (ERA & Demo tracks) – 2021, 2022, 2024
EASE: International Conference on Evaluation and Assessment in Software Engineering (Research track) - 2026
MobileSoft: International Conference on Mobile Software Engineering and Systems (Research track) - 2022, 2023
FSE: International Symposium on the Foundations of Software Engineering (Artifact track) – 2021, 2022
ASE: International Conference on Automated Software Engineering (Artifact track) – 2021, 2022
ICSE: International Conference on Software Engineering (Demo track) – 2023

Journal Reviewer

TSE: IEEE Transactions on Software Engineering – 2020, 2021, 2022, 2024, 2025
TOSEM: ACM Transactions on Software Engineering and Methodology – 2025, 2026
ESE: Empirical Software Engineering - 2021, 2022
SPE: Journal of Software: Practice and Experience – 2020

Institutional Service

CS Scholarship Committee, Member (2023 – Present)
College of Engineering Research & Graduate Committee, Member (2025 – Present)
College of Engineering Library Committee, Representative (2024 – Present)
Institutional Review Board (IRB), Alternate Member (2023 – Present)

Invited Talks

Improving Software Quality: Strategies, Techniques, and Tools for Test Evolution, IEEE Siouxland Section Computer Society Speaker Event, South Dakota State University, Dec. 2025

Awards and Honors

National Research Foundation of Korea Postdoctoral Fellowship (March 2017 – February 2020)
Postdoctoral Researcher Funding supported by the National Research Foundation of Korea

Brain Korea 21 Plus Fellowship (September 2011 – February 2017)
Graduate research assistantship support during M.S. and Ph.D. studies

KNU Honors Scholarship (September 2013 – August 2015)
Merit-based scholarship awarded by Kyungpook National University for Ph.D. study

KNU Honors Scholarship (September 2011 – February 2013)
Merit-based scholarship awarded by Kyungpook National University for M.S. study